AIP Conference Proceedings

RESEARCH ARTICLE | NOVEMBER 17 2025

Human action recognition in video data based on deep learning methods

Shaimaa H. Shaker ✉; Ammar Fakhri Mahdi; Rehab F. Hasan; Raghad Abdulaali Azeez

AIP Conf. Proc. 3395, 030020 (2025)
https://doi.org/10.1063/5.0301806

View Online
Export Citation

Articles You May Be Interested In

Targeted advertising campaign using machine learning
AIP Conf. Proc. (January 2025)

Human action behavior recognition in still images using ensemble deep learning approach
AIP Conf. Proc. (February 2025)

Human activity recognition system using smartphone based on machine learning algorithms
AIP Conf. Proc. (October 2022)

09 October 2026 04:30:20



AIP Publishing

Human Action Recognition in Video Data Based on Deep Learning Methods

Shaimaa H.Shaker [1,a)], Ammar Fakhri Mahdi [1,b)], Rehab F.Hasan [1,c)], and Raghad Abdulaali Azeez[2,d)]

[1] *Computer Science, University of Technology, Baghdad, Iraq*
[2] *Information Technology Unit, Collage of Education for Human Science-ibn rushed,University of Baghdad, Baghdad – Iraq*

[a)] *Corresponding author: Shaimaa.h.shaker@uotechnology.edu.iq*
[b)] *ammar.f.mahdi@uotechnology .edu.iq*
[c)] *Rehab.f.Hassan@uotechnology.edu.iq*
[d)] *raghad.azeez@ircoedu.uobaghdad.edu.iq*

09 October 2026 04:30:20

ABSTRACT. Recently, computer vision has made great strides in domain of human activity identification from video data, which has attracted increasing attention. To determine which human behaviors, appear in videos, action recognition is considered a must. The techniques for automating the categorization of particular human behaviors are covered in this study. This endeavor aims to assist academics who study action recognition technology with methodical knowledge acquisition, dataset selection, comprehension of present issues, and future research approaches that show promise. Depending on methods that fall into two categories: deep learning-based methods and hand-crafted features. The initial part of this work spotlights enhancing the presently efficacious hand-crafted methods. It helps the current feature detector's function. In the other part, a new combination of colorful point representations is tested to classify mortal conduct. This section's benefactions are estimated using semi-supervised literacy algorithms, which work both labeled and unlabeled samples, as part of deep literacy approaches. The parallels and differences between the approaches' attributes are emphasized. Therefore, several vision-grounded modalities are offered by the standard HAR datasets that are now accessible. The benefits and downsides of colorful infrastructures within the suggested four-position bracket are examined. Next, a list and commentary are provided for the most accurate approaches on both the most recent and well-liked datasets. This review aims to provide an overview of the modern in vision-based deep learning HAR techniques through an analysis of recent work.

Keywords: Action classification, Semi-supervised learning, hand-crafted features, Deep-learning.

INTRODUCTION

The detection of hazardous events or the monitoring of solitary individuals are only two of the many uses for human activity recognition (HAR), which is a popular area of study [1], Data collection, preprocessing, segmentation, feature extraction, and classification are the procedures that make up a conventional HAR system [2]. After then, a huge amount of restricted and local features, a movement-range independent featurealter, and a bag-of-words visualization are employed to quantify every local attribute into a vector histogram. Finally, vector-based classifiers are used in testing videos to achieve action recognition. When the movies are entirely straightforward, recognition techniques have produced positive results. Uncorrelated data may nevertheless be introduced to the bow once the local features are retrieved and quantized.In between supervised and unsupervised learning, a class of machine learning tasks and approaches called semi-supervised learning employs modest amounts of unlabeled data for training [3][4].

The 2nd International Conference on Advanced Research in Pure and Applied Science
AIP Conf. Proc. 3395, 030020-1–030020-16; https://doi.org/10.1063/5.0301806
Published under an exclusive license by AIP Publishing. 978-0-7354-5281-7/$30.00

After carefully reviewing the literature, computer vision systems based on HAR may be broadly divided into: (a) Conventional hand-crafted feature approaches, which are then afterward ainstructable classifier designed to identify actions. (b) deep learning-based techniques, which can also automatically extract features from unprocessed data [5]. In HAR applications, deep learning is getting more and more popular[6]. While deep neural networks automatically learn and extract features through a continuous minimization of a cost function, traditional pipelines involve a feature extraction phase prior to training a classifier.It's quite difficult to compile annotated films encompassing a range of actions because the existing action recognition technology uses a supervised method. A problem with overfitting would naturally arise in such a situation, when activity recognition performance is constrained and excessively complex. Modern techniques for visualization-based individual recognition are reviewed, discussed, analyzed, and compared in this paper. Significant applications, well-known benchmark datasets, and methodologies based on deep learning and handcrafted techniques are all introduced. Also, this study examined several recognition model architectures, such as fusion, modalities-based, and vision-invariationmodels [7-9]. The various architectures for for visualization-based individual recognition models are briefly described. To assist investigators in charting a course for potential future work, recent research efforts are presented and described. EsraaA. Mahareek (2023) [11],

A novel deep learning technique was used in this study to detect anomalies in surveillance videos. Using a deep network trained to recognize objects and human activity in films, the proposed method was developed. To identify anomalies in surveillance videos, the proposed method combines the advantages of ConvLSTM and 3DCNN. Measured up to modern methods, ConvLSTM has the potential to achieve high accuracy in both indoor and outdoor contexts by decreasing false positives and increasing accuracy. In [8] made a compact representation recommendation using the 2DPCA-2DLDA method and a global GIST feature. Using an SVM classifier, the activities have been identified. demonstrates excellent feature discrimination/dimension trade-off, and the action recognition accuracy is impressive. Utilize an motion history image in conjunction with optical flow to characterize restricted activities of body componentsin a large-scale chronological model. Action model training and testing are done with SVM classifiers. At a rapid processing rate of 47 frames per second on 200x150 images, both test datasets demonstrated 100% recognition rates [12].

The field of HAR and related approaches has seen the publication of numerous significant survey and review publications. Nevertheless, published reviews typically become outdated. For this reason, although being a difficult and demanding undertaking, producing an updated study on HAR is greatly needed. This review presents the latest techniques for vision-based human action recognition, along with analysis, and comparisons. Popular benchmark datasets, important applications, and methodologies based on deep learning and handcrafted techniques are presented. Additionally, this article introduces various recognition model designs, such as hybrid, modality-based, and view-invariant based models. A concise overview of various architectures for vision-based action recognition models is presented.

Section 2 presented the main concepts of human activity recognition and various dataset with few relevant papers on this topicin addition some approaches based on handcrafted action features and action features based deep learning. Section 3 introduced the semi-supervised learning algorithms. After that, sections and 5 reported some comparisons, discussion and Conclusions.

HUMAN ACTIVITY RECOGNITION

Many studies and methodologies looking at HAR have been conducted in recent years. A complex and difficult topic of research, finding patterns in HAR is still ongoing. Earlier deep learning-based HAR research is discussed in this section. Smart home, senior care, and health care are just a few of the industries where HAR can be applied. Various tools have been employed to detect behavior, including radio frequencies, cameras, and wearable sensors. Research on gathering HAR data and using sensors to identify activity has been prompted by developments in network technology [13-15]. Within the HAR system, there are two primary classifications: The first is Static cameras positioned in different locations for surveillance are part of the vision-based HAR system, which records and stores videos on servers. After that, monitoring is done with these recorded movies or camera feeds. Among the applications of this kind of HAR are traffic control, crowd surveillance, public safety, and road safety. The second is sensor-based HAR that in recent times, cellphones have emerged as a powerful instrument for international communication and human research. Smartphone cameras that are built right into the device can continuously record data about human activity. Patient monitoring systems, tracking a sports player's

09 October 2026 04:30:20

activities, and other similar applications can benefit from this kind of HAR[16].Wearing a variety of wearable sensors, participants were observed by Lara, O. D. et al., who used external datasuch as location data and environmental signals to identify the human activity of the subjects [17]. As a result, gathering information in the area of human activity recognition is challenging since it requires combining data from multiple sources, including wearables, cameras, and location data. Furthermore, to tackle the issue of sensor attachment and portability, image-based video techniques have been employed [18]. Through the use of the hidden Markov model (HMM), N.Robrtson, et al. created a method for HAR in capture on film progressions and produced managements for viewes[19]. Unfortunately, research has looked at how to use sensors installed on gadgets like smartphones to automatically detect activity because wearing or attaching wearable sensors can be uncomfortable for people [20, 21]. R. Segundo, et al. additionally identified individual action using HMM-based models that used smartphones [22].Feature learning and handcrafted feature approaches comprise the summary of action classification techniques. Many kinds of data can be processed using these techniques. incorporating interaction identification techniques, action detection techniques, and action feature representation techniques, as illustrated in fig.1. Classification and detection of human actions can be separated into two categories. The process of analyzing a fragmented video that just has one act with the intention of needs to be placed into a certain actclass is called "action classification." Each action in the movie is identified by its start and end times, location in space, and action category by action detection [23].

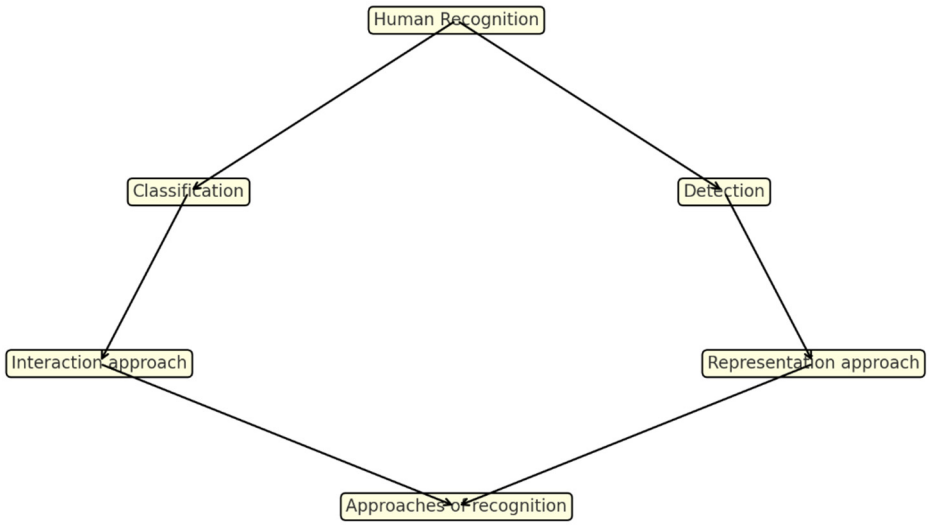


FIGURE 1: HAR ApproachesClassification

Every one of local-features are then quantized into a histogram vector after a significant amount of restricted or cramped features and movement-range invariant feature alter are retrieved from videos. After that, the action recognition in test videos is ultimately carried out by vector-based classifiers. Recognition algorithms have shown encouraging results when the movies are completely basic. Still, for the duration of the extraction and quantization of the limited characteristics, sounds and uncorrelated information could find their way into the bow. It follows that when the movies include certain camera shake, occlusion, a cluttered background, and other issues, these algorithms are typically not robust and cannot be applied broadly.Useful components of activities, for example associated entities, human signs, performance, etc., should be utilized to develop aobviouser semantic sympathetic of human activities in order to achieve recognition accuracy. Previous attempts have shown how useful it is to take advantage of related object or person stances or movements. Particularly for real-world videos, the algorithms might need to be trained on a large number of movies in order to get good performance [24]. Nevertheless, individual act depiction and identification based on hand-crafted characteristics is limited by various reasons such camera movement, occlusion, complicated situations, and the limits of individual discovery and pose approximation approaches. Deep learning techniques are more effective at automatically learning features from photos than created action features [23]. The conventional human action recognition model is depicted in Figure 2.

09 October 2026 04:30:20

Figure 3 displays the basic architecture of the deep learning-based framework for human action recognition. Four categories of benchmarks to help readers choose the best dataset for assessing their work: A description of each dataset is provided in Table1. A single-action (I), human-human interaction, human-object relations and performance (II), surveillance (III), and spectator sport movies and additional kinds (IV). Every dataset varies in complexity according on its recorded setting. Precursor benchmark datasets, like KTH [28] and Weizmann [29], for instance, were created in a laboratory setting using idealized human actions; they are all made up of uniform backgrounds and basic, unrealistic movements. Already, on these datasets, a few of techniques have very high recognition rates [27]. Ikizler et al. [30] and Brahnam et al. [31] have reported 100% accuracy on the Weizman [29] test, demonstrating the rise in performance with time to peak accuracy. Stated differently, we can assert that our action recondition systems have resolved the unrealistic datasets already. Under lab settings, a different dataset with numerous perspectives, called IXMAS, has also been created [32].

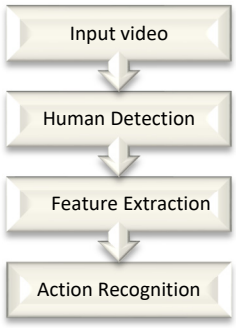


FIGURE 2: Conventional Human Action Recognition Model

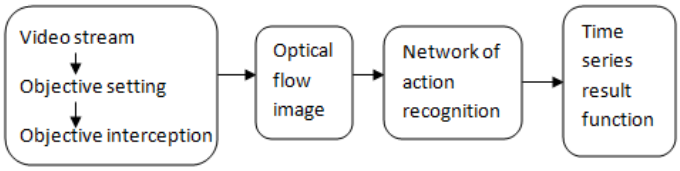


FIGURE 3. Framework of Deep Learning-Based for HAR

TABLE 1. A Description of Dataset

Author/year	Dataset(category)	No.of classes	Details
Schuldt et al. [28] 2004	KTH (I)	6	On foot, running, boxing, and wave by hand& applause.
Gorelik et al.[29] 2005.	Weizman(I)	10	Walk, run, jump, twist, and one&twohands wave, jump in place
Weinland et al. [32] 2006	IXMAS(I)	13	intersect arms, scrape head, be seated, rotate, gait, lift up
Ryoo et al. [33] 2010	UT-Interaction(II)	6	Shake-hands, indicate, hold close, push, punt.
Li et al. [34] 2010	MSR Action3D (II)	20	High & horizontal arm wave, hit, forward punch, high throw
Oh et al. [35] 2011	VIRAT 2.0(III)	12	Filling carwith objects, unlocking a car trunk
Soomro et al. [36] 2012	UCF-101(IV)	101	Horse riding, hula hoop, ice dancing, skiing, skijet, sky diving.
Karpthy et al. [32] 2014	Sports-1M(IV)	487	managing club, limit hiking, tricking, omitting cord, loose- coating

09 October 2026 04:30:20

Benchmarks that can be replicated. Publicly accessible software libraries would be used to execute well-known recommendation algorithms in this step. The findings would then be published and contrasted with those obtained for other datasets under other circumstances (such as data splitting techniques and assessment measures). Using well-known recommendation algorithms, the dataset in question will be tested in order to determine its baseline performance. Examining the consistency and durability of the findings is also made possible by repeatedly running the same tests with progressively larger dataset slices. Many assessment measures, as frequently used in the research on recommender systems (Shani and Gunawardana, 2011) [38], can be used to quantify performance outcomes, taking into account additional settings (Campos et al., 2014) [39]. Even though consistency should be the main goal, the assessment process is a variable that can be examined as well. If this isn't done, the results won't be comparable between studies [Said and Bellog 2014][40], such a technique must to, if at all feasible, be as similar to the actual situation in which the dataset will be utilized [41].

RELATED WORKS

A few relevant papers on semi-supervised and machine learning in HAR are covered in this section. A comprehensive literature assessment of HAR for production and logistics was carried out by Reining et al. in2019. A thorough review of the most recent developments in deep architectures, statistical pattern recognition, and HAR techniques is provided in this survey. There are industrial applications for this study. In 2020, Beddiar and colleagues conducted a survey on acyivity detection and identification based on visualization, grouping all of the research findings based “Handcrafted-feature” and “feature learning” approaches were covered by the writers, who also included implementation details for each technique. In a similar vein, Zhu et al. in2016 also looked at learning-based and handmade methods for action recognition.The authors first assessed the shortcomings of the handcrafted approach before briefly highlighting the development of HAR's deep learning algorithms up till 2016. Hussain et al.in 2020, examine numerous fields of HAR with the major focal point on device-free approaches, namely RFID. The literatureoffers a classification that is divided into three subcategories: interaction-, motion-, and action-based. These domains are further broken down into ten subtopics, each of which has the most recent HAR methodologies given. The authors of this study, Sh.Vijeta et al. in2022, hope to provide a complete comparative assessment of individual action detection and identification using visualization, with a particular emphasis on deep-learning techniques across a range of benchmark video datasets. Next To illustrate the development of HAR techniques, a brief comparison is also presented between the handcrafted feature-based approach and its merger with deep learning [25].Machine learning models using hand-crafted features perform well in early HAR research. Mobile devices' numerous sensors provide raw sensor data. Using domain expertise, handcrafted features are created from this data. Machine learning models, like random forest, naive Bayes, or SVMs, are trained and utilized in HAR using the manually created features. Handcrafted feature design calls for domain expertise. To represent the time series signals in HAR, it is therefore advantageous to create a methodical feature learning tactic. The emergence of deep neural networks (DNNs) as feature extraction techniques for HAR has made and they have greatly advanced in numerous fields. The unsupervised feature extraction is investigated in the first HAR deep learning method [42]. It performs better than statistical features and principal component analysis. Following then, the popularity of convolutional neural networks (CNNs) was attributed to its translation invariance and locality preservation. While a 2D CNN treats multichannel sensor readings and views the image as collection of signals, a 1D CNN models sensor modality [43]. Promising performance in HAR has been demonstrated by deep recurrent networks, particularly long short-term memory cells (LSTMs), in capturing the chronological relationships of data-sensor [44]. But in order to prevent overfitting, LSTMs require a large amount of labeled data because of their complexity. Both labeled and unlabeled data are used to train the model in semi-supervised learning. For HAR, an online adaptation technique is suggested for semisupervised learning [45]. The no-labeled data are choosyappended to the training set and iteratively annotated using the self-learning based methodologies [46]. By connecting labeled and unlabeled data, the graph-based technique creates several graphs that propagate labels based on feature similarity[47]. These methods, however, handle categorization and label propagation as two distinct processes. Recently, ladder networks [48], a semi-supervised technique, can simultaneously train a neural network on a labeled dataset and a deep auto-encoder on an unlabeled dataset. Since the CNN offers steady dormant representations at every net stage, which conserved locality, J. Yang, M. Nguyen et al. incorporated it in [49].For supervised

09 October 2026 04:30:20

and semi-supervised learning algorithms, we used the many-sensor and CNN construction [50].Table 2 summarizes the performance of some related works.

TABE 2. Summary of Some Related Works

Author/Year	Focus of study	Method	Performance
Reining et al. [2019]	systematic literature review of HAR for production and logistics	review of developments in deep architectures, statistical recognition& HAR techniques	beneficial for industrial applications
Beddiar et al. [2020]	various techniques of Handcrafted-feature and feature learning-based approach,	conducted a survey on activity detection and identification based on visualization, grouping the findings based "Handcrafted-feature" and "feature learning" approaches	highlight related literature based on human activity types – Elementary human actions, Gestures,Behaviors, interactions, group actions, and events, which advocate HAR approaches at the minute level.
Hussain et al. [2020].	focus on HAR with the major focal point on device-free techniques, especially RFID	offers a classification that is divided into 3 subcategories: interaction-, motion-,& action-based. These domains are further broken down into10 sub-topics, each has the recent HAR methodologies given.	Compare the 10 sub-topics according some benchmark datasets
Sh.Vijeta et al. [2022]	focus on deep learning techniques on various benchmark video datasets comprehensively provide a comparative of individual action detection &identification using visualization, with a deep-learning techniques and a brief comparison is between the handcrafted feature-based approach and its merger with deep learning	propose a new taxonomy for categorizing the literature as CNN and RNN-based approaches. Plus divide these approaches into 4 sub-classes with methodologies and theirexperimental datasets	short comparison was with the handcrafted feature-based approach and its fusion with deep learning
R.Antti eat al.[2023]	combine supervised learning with unsupervised learning in deep neural networks	By connecting labeled and unlabeled data, the graph-based technique creates several graphs that propagate labels based on feature similarity	resulting model reaches modern performance in semi-supervised MNIST & CIFAR-10 classification, in addition to permutation-invariant MNIST classification with all labels
M.Stikic et al [2005]	Foucs on Multi-graph based semi-supervised learning for activity recognition	propagates information through a graph that contains both labeled and unlabeled data. By combining multiple graphs based on feature similarity and time.	approach outperforms multi-instance learning approach and in some cases even outperforms fully supervised approaches.
J. Yang, M. Nguyen et al. [2015]	focuses on HAR problem, inputs are multichannel time series signals acquired from a set of bodyworn inertial sensors and outputs are predefined human activities	propose a systematic feature learning method for HAR problem. This method adopts a deep convolutional neural networks (CNN) to automate feature learning from the raw inputs in a systematic way	learned features are deemed as the higher level abstract representation of low level raw time series signals. Unified in 1model, feature learning

09 October 2026 04:30:20

Z.Mingst al. [2017]	present 2 semi-supervised methods based on CNNs to learn discriminative hidden features. For supervised & semi-supervised learning algorithms, used many-sensor & CNN- construction	Our semi-supervised CNNs learn from both labeled and unlabeled data while also performing feature learning on raw sensor data	and classification are mutually enhanced workss on 3 real world datasets, then have CNNs outperform supervised methods and traditional semi-supervised learning methods by up to 18% in mean F1-score (Fm).

HANDCRAFTED ACTION FEATURES

When there is a shortage of labeled data, feature learning a technique allows us to potentially enhance HAR performance by modifying the feature representations of the labeled data using unlabeled data. On the other hand, the majority of earlier semi-supervised HAR techniques depend on manually created characteristics. Because there is little chance for attribute learning with the no-labeled data, the usefulness of the no-abeled data is restricted when features are produced [50].The purpose of handcrafted features is to capture the motions of people as well as the temporal and spatial changes that occur in action-representing videos. To identify action, these variables are primarily utilized in traditional machine learning techniques as Boost, SVM, and likelihood chart style.This was the first method for recognising human actions. The main techniques were improved by A.Bobick,J.Dvis [51], suggested using movement energy images and movement record images to represent activity; Zhang et al. [52], divided the mid district of the person mass-body in movement record images using polar coordinates and used a Motion Context descriptor based on SIFT to represent the activity; Marszaleet,A.Klasr,et al. [53], proposed using the 3DHOG feature of activities videos;soSomsundaram et al. [54], computed the similarityof self of a video in the varitydimensions, so offering a depiction of human behavior. In order to identify the important areas of a video and depict human behaviors, Yousa,S.Nazir et al. [55] combined the 3D-Harris spatio-temporal features and 3D scale-invariationattribute transform finding techniques. Ni and others. RGB and depth data's spatiotemporal properties were combined by Jalal et al. [56]. For both online and offline human-object interaction recognition, Meng et al. suggested a spatiotemporal modeling of human-object interaction movies [57]. This modeling made use of both local depth information and skeleton data. Action Recognition approaches utilizing traditional machine learning algorithms are depicted in Fig.4.

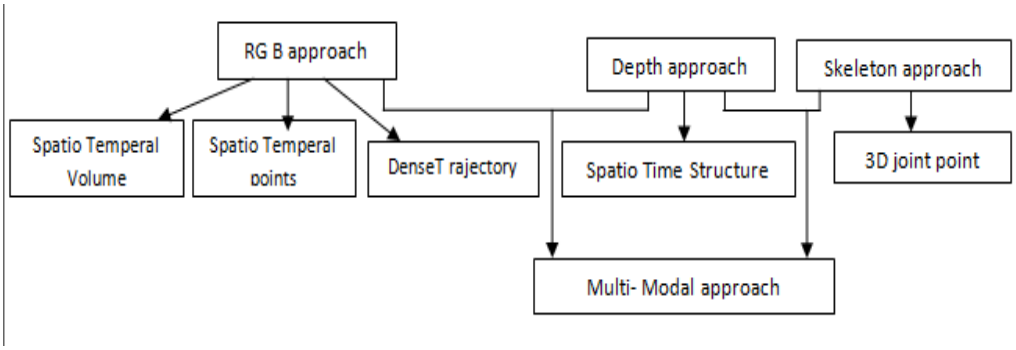


FIGURE 4. Action Recognition Approaches Based on Handcrafted Features Utilizing Machine Learning

ACTION FEATURE REPRESENTATION METHODS BASED ON DEEP LEARNING

Deep learning-based models have been used in HAR more frequently lately [59–61]. Current methods are mostly founded on research employing convolution neural-networks (CNNs)[62], long short-term memory (LSTM)[63], deep neural networks (DNNs), plus other techniques. Wu.L.Zhanget al. integrated HMM with DNN to examine HAR as deep learning in progress to be deliberate in the domain of

09 October 2026 04:30:20

action detection and identification [64]. In order to compare the functioning of artificial_ neural_ networks, SVM, and deep_belief networks (DBNs), authorsobtained the distinctiveness of data from Smartphone sensor[65]. They demonstrated that deep_belief networks achieved the most excellent for behavior detection. For identification of person action, L. and S.Wanet al. evaluated each of many models like SVM, CNN, LSTM, bidirectional LSTM, and MLP. Then the results showed that CNN models performed best on the HCIHAR and PAMAP2 datasets [66]. Using the HCI-HAR dataset, KIM and SUllah, et al. demonstrated a frivolous FCN-LSTM representation [67]. These studies have simply made an effort to document modifications to sensors or adjustments to discriminant models for the identification of human activities. Information that can be employd to provide fine functioning, even for modest numbers, is referred to as valuable data. Eventually, this may result in fewer labeling duties. Research is being done in this regard using a variety of techniques[68].

To make predictions, the HAR-Net combines high-level data taken from a convolutional network with manually created features. It was shown that the suggested method performed 0.9persentage improvedthe original MC-SVM technique. UCI dataset experiment findings show that the lack of conventional characteristic engineering and D-learning techniques can be compensated for by combining the two types of features. A network architecture termed HAR-Net for the wearable sensor-based human activity detection challenge. In Fig.5, the network architecture is shown. The separative convolutional layer will process the enteryof the net, which will have nine channels and a measurement of (1, 128, 9). This indicates that the identical filters will convolve each of the input's nine channels in turn. The input's channels will each be treated concurrently by filters with (1, 1), (5, 1), (9, 1), (13, 1), (17, 1), (21, 1), (25, 1), (29, 1), and (33, 1) in that order, everygo after by a layer called max-pooling hasdimention(11, 1). The above-described prototype of a layer go aftera max-pooling layer which repetitive 4epoch in a row. The concatenation of hand-crafted features, including statistical characteristics in different domain, that done with the production that has been flattened out from the convolutional layers. then fed into many-neurons with relu_activationtask in the initial completely linkedlevel.The first completely linked layer's output that fed into the next level. To categorize various types of activities, the output of the next fully connected level is subjected to Softmax. To the HAR-Net using the defaulting constraint of Adam-optimizer[69].In order to address Human Activity Recognition, a novel strategy leveraging Inception Block for neural networks with manually created attributes was put forth. Convolutional neural networks, or HAR-Net, present an enticing solution for implementing HAR in the healthcare sector and protecting vulnerable populations. Convolutional neural networks have the advantages of scale invariance and local dependency [70], which makes them a viable option for use in the HAR sector.

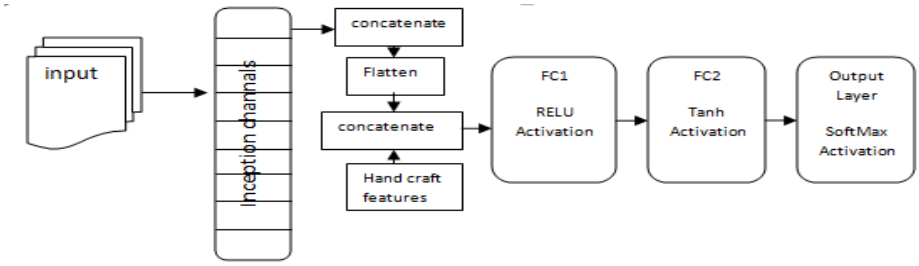


FIGURE 5. HAR-Net Architecture

Because deep learning extracts additional self-adaptive features from the model, HAR-Net outperforms the conventional MC-SVM method in terms of prediction accuracy. Using the same public domain data set, the study demonstrated that the combination of deep learning techniques with conventional feature engineering can outperform MC-SVM methodology in terms of prediction accuracy [69].

SEMI-SUPERVISED LEARNING

In order to predict labels, deep learning approaches use mathematical operations on the worthsof the enter data and model weights. A method called supervised learning (SL) uses labels and data to learn in a variety of domains, including regression and classification. Large-scale classification and regression problems are commonly handled by SL, which uses labeled data to learn and make use of it. Supervised learning was a major machine learning axis since before time ML was employed to categorize specific

09 October 2026 04:30:20

information or forecast rates. To train SL, however, thousands to millions of learning data points may be required, and labels are necessary for every data point. Because of these issues, SL is not useful in the absence of sufficient learning data. A method called semi-supervised learning, or semi-SL, was created to make up for these shortfalls[71]. Refer to fig.6. It is a method which is employed to label and nolabeled data with basic criteria and to train ML models again by previous learning data to improve functioning, based on the anticipated values of the learned model, as illustrated in fig.7.Table3 provides a detailed description of the various designs used in databases. versions six and eightutilized the 2D ViTstructure with a batch of sixteen, while versions 4 & 5 utilize the 2D ResNet 50 as the foundation structure with a batch equivalent to 8. The classifier employed temporal grouping and LSTM to adjust the 2D network to the 3D situation. Thirty epochs, a cut_middleon images measuring 224*224pixels, arbitrary flat inversion, scaling values,normalization, change to tensors, and arbitrary chronological assortment were the parameters used for the runs[72]

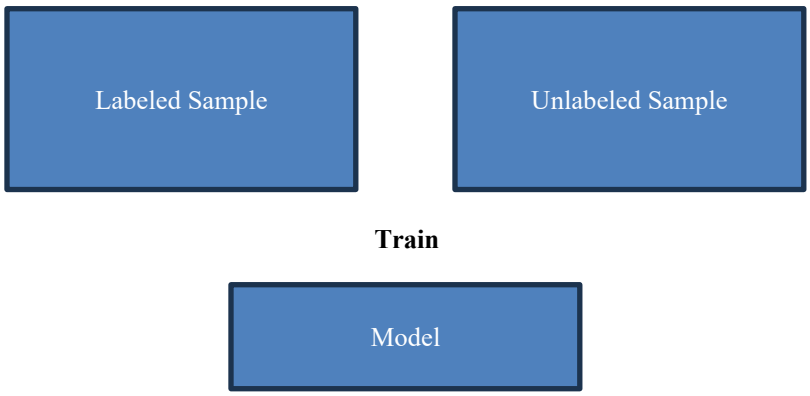


Figure 6. Semi-SL

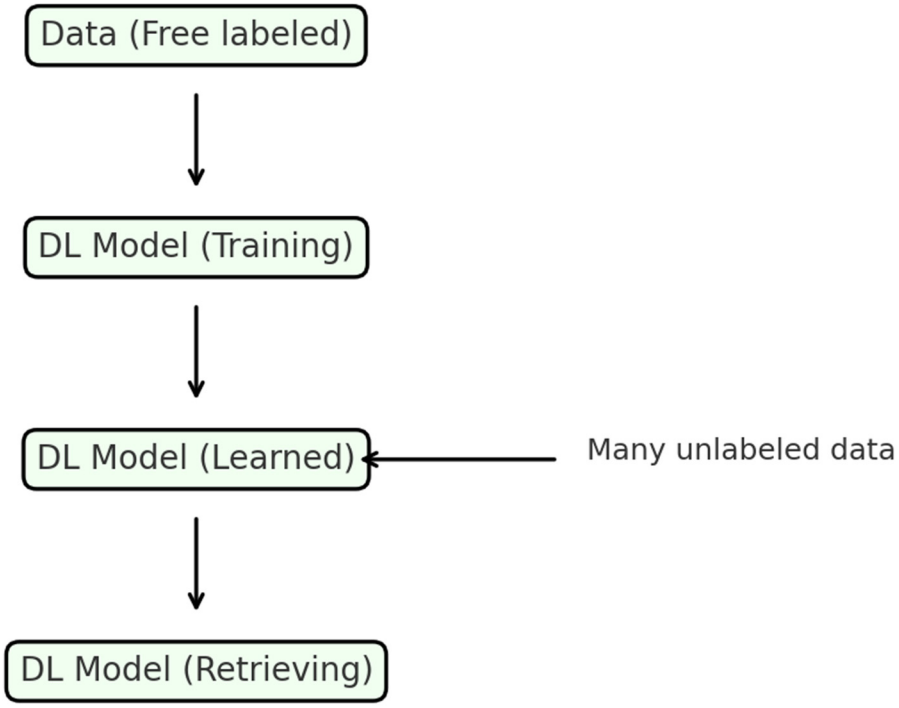


Figure 7. Semi-SL architecture

09 October 2026 04:30:20

TABLE 3. Some Architectures Using a Semi-Supervised Learning Technique on HMDB51.

Version	Classifier	Rate of learning	Time	Approach
DINO4	2× FCN	0.1	1	Temporal Grouping
DINO5	1× FCN	0.1	1	Temporal Grouping
DINO6	1× FCN	0.1	1	Temporal Grouping
DINO7	LSTM+1*FCN	0.1	4	2-D with LSTM
DINO8	LSTM+1*FCN	0.1	1	2-D with LSTM

Haixia et al in[74], create a deep HAR model by combining semi-supervised and active learning into a single framework. This enhances the model's functioning in little explanation regimes by choosing the most usefultrials for annotation and utilizing the data from large amounts of unlabelled instances. As far as ,this is the first attempt to integrate these two methods into a single HAR framework. In order to use the chronological collection of the DL model depending on constancy regularization, a fresh unsupervised loss term is implemented. This effectively allows SS-Learning in conjunction with the supervised loss component. With the help of this unsupervised loss term, prediction uncertainty is lessened, resulting in more precise and reliable forecasts for activities. seefig.8. The steps of deep model are:

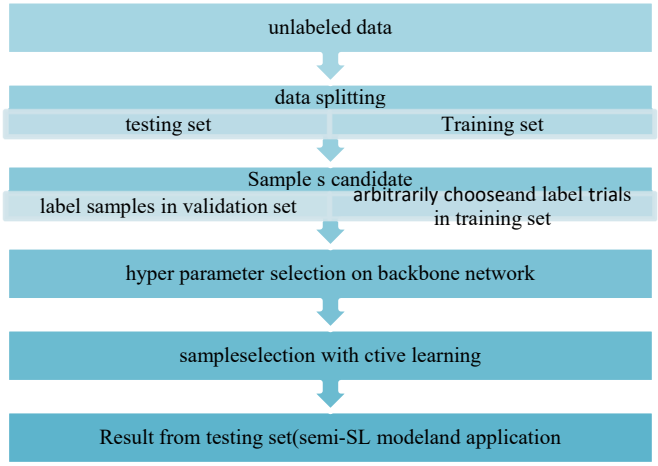


FIGURE 8. Model Workflow.

First step: Sample selection based on active learning. This is an iterative process in which we use preexisting labels to train a SD- CNN model. Next, predict the unlabelled candidates and actively select the mainly usefultrails to apply for new explanation, resulting in an efficient set of labels. Second step : Temporal ensembling is used in semi-supervised model training. Dropout regularization is used to feed the network with both labelled and unlabelled samples. Fig.9 illustrates the fowchart of the approach. The following findings from thorough comparative experiments: Effective trail choice and the use of large amounts of unlabeled data provide modern performance and a significant reduction in the annotation cost when active learning and semi-supervised learning are combined. So the efficiency of collection harmony in boosting the resilience of HAR models is demonstrated by the statistical importance and cross validation tests. [74]

DISSCUSION

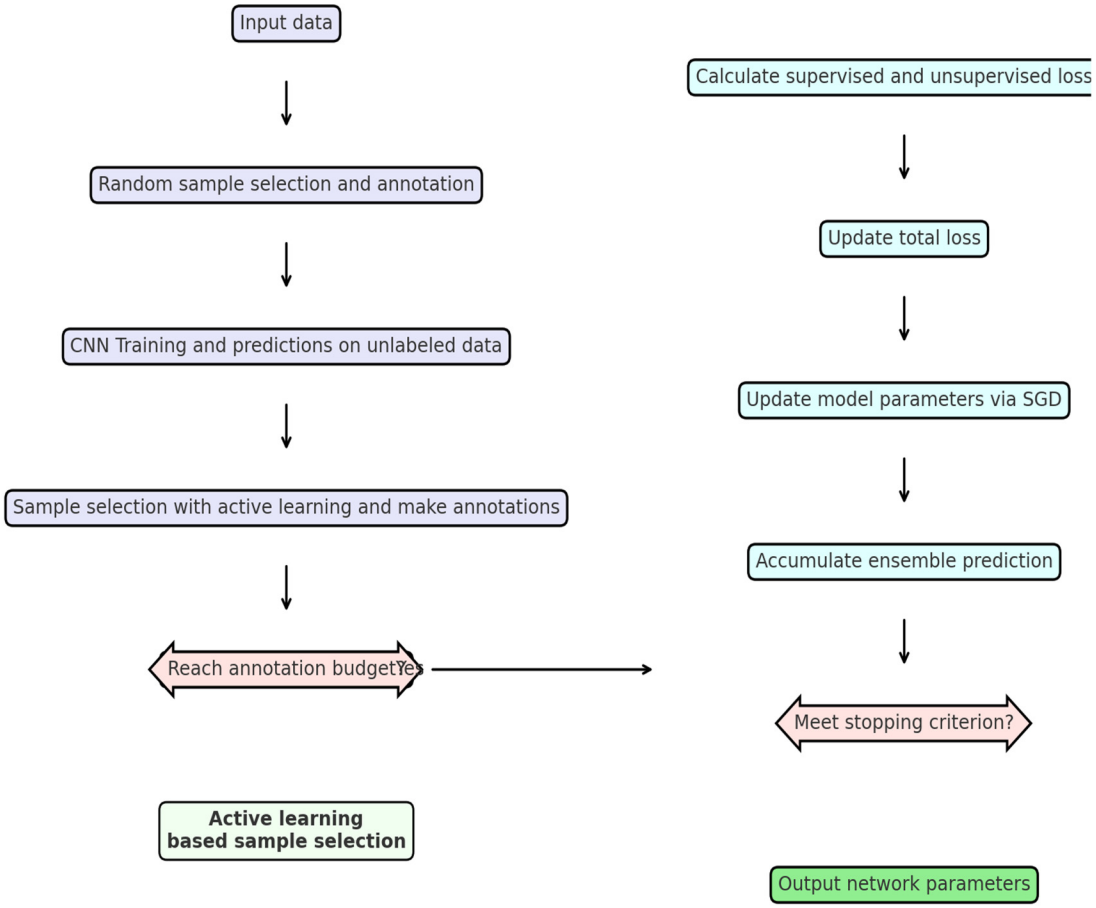
TABLE 4, Comparison of Types for Human-Action Detection and Identification [75–78].

	Characteristics	Hand-Crafted Feature Based Models	Deep Learning Based Models
1	Inter-class and Intra-class	ineffective handling of these issues.	From such models, hierarchical and translational invariant

09 October 2026 04:30:20

			characteristics are derived to address this issue.
2	Feature extraction and Representation	Need some detector algorithms to extract efficient patterns to get recognition accuracy	be trained features directly from data
3	Generalisation plus Diversity	using of techniques for FSand DR that are not very generalizable.	Discover spatial, temporal and scale, invariant elements from data
4	Data preparation	require data pre-processing and normalization to get high performance.	Optional pre-processing and normalization to get high performance

One technique that potentially leverage unlabeled data to boost learning performance is semi-supervised learning. Noteworthy is the fact that in reality labeled data applications are typically challenging or costly to acquire since they need the work of annotators who are human. But it's simpler to obtain unlabeled data because human labeling is not necessary work. The potential to leverage the free unlabeled data makes semi-supervised learning a popular approach subjects related to machine learning. Table4explains a difference of modelsforf HAR. So, Table5 shows some subsistence advanced techniques using deep learning models and various datasets for human action recognition[5].



FIGUER 9. Flowchart of the Network approach

09 October 2026 04:30:20

TABLE 5. Techniques Using DL Models Plus Data-Sets for HAR

	Approach	# Design	Dataset	performance
1	Co-op model /2018	CNN	NTU-RGB+D	86.42
2	FWMDMM /2019	CNN	MSR-Daily Action	92.90
3	MB_MHI/2020	Res-Net	MUHA-Vi	83.80
4	MV-RAMDMM/2020	3D-CNN	MSR-Daily Action	87.50
5	S-CNN/2020	Res-Net	UCF101	89.00
6	S-CNN/2020	VGG-16	UCF101	82.58

One possible way to improve the feature extraction technique is to create a hybrid network structure that combines the best aspects of current deep learning-based techniques. Every algorithm is thought to have been created with a particular focus on feature extraction and learning. Consequently, rather than creating a technique for a specific dataset, limited area, or application, fine-tuning a few well-known methods can be helpful in a variety of HAR applications. In addition, this hybrid network might be useful in addressing the major problems associated with HAR, like occlusions, crowded backgrounds, and camera rotation angles. tested algorithms on different volumes of labeled and unlabeled data in order to determine how useful they were. An approach has been put out to improve the performance of action recognition through the use of deep learning technology. This study summarizes deep learning and divides it into two groups: supervised and unsupervised models. Nonetheless, because of their extensive capability and exceptional efficacy in executing recognition systems, supervised models are taken into consideration in this work.

CONCLUSION

Our aim in doing this research is to provide readers with an in-depth understanding of the current situation of DL-approaches used to recognize behavior in videos, as well as the development process in general. Many related works have offered a methodical analysis of a number of deep learning architectures and their applications in activity recognition and associated jobs. This research introduce some comparisons of theaccuracy of identification for DL-approaches with alternative tacticeshave demonstrated that deep learning is now the most effective option for both classifying and identifying human behavior and action. Furthermore, an analysis has been conducted on the features of the majority significant DL-models for activity recognition in order to present current tendency and unresolved subjectes for further research in this domain. This paper will aid involved readers in choosing approximate procedures and data-sets to make new results as long as a catalog of datasets with varying levels of complexity. Even while DL models have advanced significantly in recent years, creating vision-based action detection systems and implementing them to improve people's lives still presents numerous obstacles. We continue to anticipate novel deep learning (DL) techniques that will enhance recognition system performance while lowering computational costs and using fewer labeled data sets. We believe that researchers working in this topic will find this survey useful.

09 October 2026 04:30:20

ACKNOWLEDGMENTS

The authors would like to thank the University of Technology, Baghdad-Iraq, and University of Baghdad, Baghdad-Iraq for the support in the present work.

REFERENCES

1. O. C. Ann and L. B. Theng, “Human activity recognition: a review,” Proceedings of the IEEE International Conference on Control System, Computing and Engineering, pp. 389–393, IEEE, Batu Ferringhi, Malaysia, November 2014
2. F. Li, K.Shirahama,M.Nisar, L.Köping, M.Grzegorzek,”Comparison of feature learning methods for human activity recognition using wearable sensors”, Sensors 2018, 18, 679.
3. H. Yahong, Y.Yi, M.Zhigang, Y.Yan, S.Nicu, Z.Xiaofang, “Semi-Supervised Feature Selection via Spline Regression for Video Semantic Recognition”, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, February 2015.

4. L. Shao, L. Liu, and M. Yu, “Kernelized multiview projection for robust action recognition”, International Journal of Computer Vision, October 2015.
5. M. Al-Faris, J. Chiverton, D.Ndzi , and A. Isam Ahmed, “ A Review on Computer Vision-Based Methods for Human Action Recognition”, Journal of Imaging 2020, 6, 46; doi:10.3390/jimaging6060046.
6. J. Wang,Y.Chen, S.Hao, X.Peng, L.Hu, “Deep learning for sensor-based activity recognition: A survey”, Pattern Recognit. Lett. 2019, 119, 3–11
7. Abbas, S.F., Shaker, S.H., Abdullatif, F.A.,”Masked Face Identification and Tracking Using Deep Learning: A Review”, Journal of Computer Science, 2023, 19(12), pp. 1423–1437
8. Abbas, S.F., Shaker, S.H., Abdullatif, F.A.,”YOLOv8-AS: Masked Face Detection and Tracking Based on YOLOv8 with Attention Mechanism Model”,Communications in Computer and Information Science, 2024, 2096 CCIS, pp. 267–275
9.]Majeed, A.W., Shaker, S.H., Saeid, A.A.,”A Real Time Face Recognition and Tracking Framework Using Lightweight Convolutional Neural Network”,BIO Web of Conferences, 2024, 97, 00029.
10. E. Mahareek, E. El-Sayed, N. El-Desouky, and K..ElDahshan, “Detecting anomalies in security cameras with 3DCNN and ConvLSTM,” pp. 0–17, 2023.
11. Y. Wang, Y. Li, and X. Ji, “Human action recognition using compact global descriptors derived from 2DPCA-2DLDA”, IEEE International Conference on Computer and Information Technology, IEEE, 2014, pp. 182–186
12. D. Tsai, W. Chiu, and M. Lee, “Optical flow-motion history image (OF-MHI) for action recognition,” Signal Image Video Process, vol. 9, no. 8, pp. 1897–1906, 2015
13. O. Seungmin, A. Ashiquzzaman , L. Dongsu, K. Yeonggwang and K. Jinsul, “Study on Human Activity Recognition Using Semi-Supervised Active Transfer Learning”, Sensors 2021, 21, 2760.
14. A. ltalbi, A.A.H., Shaker, S.H., Ali, A.E,”Anomaly Detection from Crowded Video by Convolutional Neural Network and Descriptors Algorithm: Survey”,International journal of online and biomedical engineering 2023, 19(7), pp. 4–25.
15. A. ltalbi, A.A.H., Shaker, S.H., Ali, A.E.,”Localization of Strangeness for Real Time Video in Crowd Activity Using Optical Flow and Entropy”,International journal of online and biomedical engineering, 2023, 19(7), pp. 52–68.
16. Htike, K. K., O. O. Khalifa, H. AdibahMohd Ramli, and M. A. M. Abushariah,“Human Activity Recognition for Video Surveillance Using Sequences of Postures.” ,3rd International Conference on E-Technologies and Networks for Development, ICeND 2014, Beirut, Lebanon, 79–82.
17. O. Lara, M.Labrador, “A Survey on Human Activity Recognition using Wearable Sensors.” IEEE Commun. Surv. Tutor. 2013, 15, 1192–1209.
18. S. Ke, H.Thuc, Y.Lee, J.Hwang, J.Yoo,K.Choi,“A Review on Video-Based Human Activity Recognition”, Computers 2013, 2, 88–131.
19. N. Robertson,I.Reid,”A general method for human activity recognition in video”, Comput. Vis. Image Underst. 2006, 104, 232–248.
20. D. Anguita, A.Ghio, L.Oneto,X.Parra, J.Reyes-Ortiz,”A public domain dataset for human activity recognition using smartphones”, Proceedings of the European Symposium on Artificial Neural Networks, Computational Intelligence and Ma-chine Learning, Bruges, Belgium, 24–26 April 2013; pp. 437–442.
21. A.Bayat,M.Pomplun, D.Tran, ,”A Study on Human Activity Recognition Using Accelerometer Data from Smartphones”,. Procedia Comput. Sci. 2014, 34, 450–457.
22. R. San-Segundo,J.Echeverry-Correa,C.Salamea, J.Pardo, “Human activity monitoring based on hidden Markov models using a smartphone”, IEEE Instrum. Meas. Mag. 2016, 19, 27–31.
23. Z. Hong-Bo, Z.Yi-Xiang, Z.Bineng, L.Qing, Y.Lijie, D.Ji-Xiang, and Ch.Duan-Sheng, “A Comprehensive Survey of Vision-Based Human Action Recognition Methods: Review”,Sensors 2019, 19, 1005.
24. M. Saniya, V. Dileep, “Enhancement of Video Action Recognition using Semi-Supervised Learning”, International Journal of Engineering Research & Technology (IJERT) ,Vol. 8 Issue 07, July-2019
25. Sh. Vijeta, G. Manjari, K.P.Anil, M.Deepti& K. Ajai , “A Review of Deep Learning-based Human Activity Recognition on Benchmark Video Datasets “,APPLIED ARTIFICIAL INTELLIGENCE 2022, VOL. 36, NO. 1, e2093705 (2901 pages).

09 October 2026 04:30:20

26. W. Zhipan, and D.Huaying," Research on Human Action Feature Detection and Recognition Algorithm Based on Deep Learning ",Hindawi, Mobile Information Systems Volume 2022, Article ID 4652946, 12 pages.
27. H. Hieu,KH.Louahdi, C.Alain,Z.Pablo, and A.Sergio," Video-based Human Action Recognition using Deep Learning: A Review", arXiv:2208.03775v1 [cs.CV] 7 Aug 2022.
28. C. Schuldt, I. Laptev, and B. Caputo, "Recognizing human actions: a local SVM approach," in Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004., vol. 3, Aug 2004, pp. 32–36 Vol.3.
29. L. Gorelick, M. Blank, E. Shechtman, M. Irani, and R. Basri, "Actions as space-time shapes," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 29, no. 12, pp. 2247–2253, Dec 2007
30. N. Ikizler and P. D. Sahin, "Human action recognition using distribution of oriented rectangular patches," in Workshop on Human Motion, 2007.
31. S. Brahnam and L. Nanni, "High performance set of features for human action classification," in IPCV, 2009.
32. D. Weinland, R. Ronfard, and E. Boyer, "Free viewpoint action recognition using motion history volumes," Computer Vision and Image Understanding, vol. 104, no. 2-3, pp. 249–257, 2006.
33. M. Ryoo and J. Aggarwal, "UT-Interaction Dataset, ICPR contest on Semantic Description of Human Activities (SDHA)," http://cvrc.ece.utexas.edu/SDHA2010/Human Interaction.html, 2010.
34. W. Li, Z. Zhang, and Z. Liu, "Action recognition based on a bag of 3D points," IEEE Computer Society Conference on Computer Vision and Pattern Recognition-Workshops. IEEE, 2010, pp. 9–14.
35. S. Oh, A. Hoogs, A. Perera, N. Cuntoor, C. C. Chen, J. T. Lee, S. Mukherjee, J. K. Aggarwal, H. Lee, L. Davis, E. Swears, X. Wang, Q. Ji, K. Reddy, M. Shah, C. Vondrick, H. Pirsiavash, D. Ramanan, J. Yuen, A. Torralba, B. Song, A. Fong, A. RoyChowdhury, and M. Desai, "A largescale benchmark dataset for event recognition in surveillance video", 8th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS), Aug 2011, pp. 527–528.
36. K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A dataset of 101 human actions classes from videos in the wild," CoRR, vol. abs/1212.0402, 2012.
37. A. Karpathy, G. Toderici, S. Shetty, T. Leung, R. Sukthankar, and L. Fei-Fei, "Large-scale video classification with convolutional neural networks," in 2014 IEEE Conference on Computer Vision and Pattern Recognition, 2014.
38. G. Shani and A. Gunawardana. "Evaluating recommendation systems. In Recommender Systems", Handbook. 257–297,2011.
39. G. Pedro, D.Fernando,andC.Ivan," Time-aware recommender systems: a comprehensive survey and analysis of existing evaluation protocols", User Modeling and User-adapted Interaction 24, 1-2 (2014), 67–119, 2014.
40. A. Said and A. Bellog ," Comparative recommender system evaluation: benchmarking recommendation frameworks", In Eighth ACM Conference on Recommender Systems, RecSys '14, Foster City, Silicon Valley, CA, USA - October 06 - 10, 2014. 129–136.
41. S.DOOMS, A.BELLOGIN, T.DE PESSEMIER, and L. MARTENS," A Framework for Dataset Benchmarking and its Application to a New Movie Rating Dataset", ACM Transactions on Embedded Computing Systems, Vol. V, No. N, Article A, Publication date: January YYYY, DOI:http://dx.doi.org/10.1145/0000000.0000000.
42. T. Plötz, N. Y. Hammerla, and P. Olivier, "Feature learning for activity recognition in ubiquitous computing," in IJCAI, vol. 22, no. 1, 2011, pp. 1729–1734.
43. M. Zeng, L. T. Nguyen, B. Yu, O. J. Mengshoel, J. Zhu, P. Wu, and J. Zhang, "Convolutional neural networks for human activity recognition using mobile sensors," in MobiCASE, 2014, pp. 197–205.
44. Y. Guan and T. Ploetz, "Ensembles of deep lstm learners for activity recognition using wearables," arXiv preprint arXiv:1703.09370, 2017.
45. B. Cvetkovic, B. Kaluza, M. Luštrek, and M. Gams, "Semisupervised learning for adaptation of human activity recognition classifier to the user," in IJCAI, 2011, pp. 24–29.
46. A. Lopes, J. Mendes-Moreira, and J. Gama, "Semi-supervised learning: predicting activities in android environment," in Workshop on Ubiquitous Data Mining, 2012, p. 38.
47. A. Rasmus, M. Berglund, M. Honkala, H. Valpola, and T. Raiko, "Semi-supervised learning with ladder networks," in NIPS, 2015, pp. 3546–3554.
48. M. Stikic, D. Larlus, and B. Schiele, "Multi-graph based semi-supervised learning for activity recognition," in ISWC, 2009, pp. 85–92.

09 October 2026 04:30:20

49. J. Yang, M. Nguyen, P. San, X. Li, and S. Krishnaswamy, “Deep convolutional neural networks on multichannel time series for human activity recognition,” in IJCAI, 2015, pp. 3995–4001.
50. Z. Ming, Y.Tong, W.Xiao, N.Le, J.Mengshoel, and L.Ian, “Semi-Supervised Convolutional Neural Networks for Human Activity Recognition”, arXiv:1801.07827v1,22 Jan 2018
51. A. Bobick, J.Davis,”The recognition of human movement using temporal templates”, IEEE Trans. Pattern Anal. 2001, 23, 257–267.
52. I. Zhang, Y.Hu, S.Chan,L.Chia,”Motion Context: A New Representation for Human Action Recognition”. Proceedings of European Conference on Computer Vision; Springer: Berlin/ Germany, 2008; pp. 817–829.
53. A. Klaser, M.Marszalek, C.Schmid, ”A Spatio-Temporal Descriptor Based on 3D-Gradients”, Proceedings of the British Machine Vision Conference, Leeds, UK, 1–4 September 2008; pp. 1–10.
54. G. Somasundaram, A.Cherian, V.Morellas, N.Papanikolopoulos, “Action recognition using global spatio-temporal features derived from sparse representations”,Comput. Vis. Image Underst. 2014, 123, 1–13.
55. S. Nazir, M.Yousaf,S.Velastin, “ Evaluating a bag-of-visual features approach using spatio-temporal features for action recognition”,Comput. Electr. Eng. 2018.
56. A. Jalal, Y.Kim, Y.Kim, S.Kamal, D.Kim, “Robust human activity recognition from depth video using spatiotemporal multi-fused features”, Pattern Recognition. 2017, 61, 295–308.
57. M. Meng , D.Hassen, and B.Jacques, “Distances evolution analysis for online and off-line human object interaction recognition” , Image and Vision Computing ,72 ,(2018), 3 .
58. B. Muhammad, Ch.Douglas, “RGB-D Data-Based Action Recognition: A Review “, Sensors 2021, 21, 4246.
59.] A. Murad, J.Pyun,”Deep Recurrent Neural Networks for Human Activity Recognition”. Sensors 2017, 17, 2556.
60. J. Yang,M.Nguyen,P.San, X. Li, S.Krishnaswamy, “Deep convolutional neural networks on multichannel time series for human activity recognition”. In Proceedings of the 24th International Conference on Artificial Intelligence (IJCAI 15), Buenos Aires, Argentina, 25–31 July 2015.
61. C. Ronao, S.Cho,”Human activity recognition with smartphone sensors using deep learning neural networks”, Expert Syst. Appl. 2016, 59, 235–244.
62. A. Al-Saffar, H.Tao,M.Talab,“Review of deep convolution neural network in image classification”, In Proceedings of the 2017 International Conference on Radar, Antenna, Microwave, Electronics, and Telecommunications (ICRAMET), Jakarta, Indonesia, 23–24 October 2017; pp. 26–31.
63. S. Hochreiter, J.Schmidhuber, “Short-Term Memory”, Neural Comput. 1997, 9, 1735–1780.
64. L. Zhang, X.Wu, “Human activity recognition with HMM-DNN model”, In Proceedings of the 2015 IEEE 14th International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC), Beijing, China, 6–8 July 2015; pp. 192–197.
65. M. Hassan, Z.Uddin,A.Mohamed, A..Almogren, “A robust human activity recognition system using smartphone sensors and deep learning. Future Gener”,Comput. Syst. 2018, 81, 307–313.
66. S. Wan, L.Qi, X.Xu, C.Tong, Z. Gu, “Deep Learning Models for Real-time Human Activity Recognition with Smartphones”, Mob. Netw. Appl. 2019, 25, 743–755.
67. S.Ullah, D.Kim, “Sparse Feature Learning for Human Activity Recognition”, Proceedings of the IEEE International Conference on Big Data and Smart Computing), Jeju, Korean, 17–20 January 2021; pp. 309–312.
68. O. Seungmin, A.Akm, L.Dongsu, K.Yeonggwang, and K.Jinsul, ” Study on Human Activity Recognition
69. Using Semi-Supervised Active Transfer Learning “. Sensors 2021, 21, 2760. https://doi.org/10.3390/s21082760
70. D. Mingtao,H.Jindong,” HAR-Net:Fusing Deep Representation and Hand-crafted Features for Human Activity Recognition”, 2018,arXiv:open access.,
71. M. Zeng, T.Le, B.Yu, O.Mengshoel, J.Zhu,P.Wu,“Convolutional Neural Networks for Human Activity Recognition using Mobile Sensors”,International Conference on Mobile Computing, Applications and Services, pp.197-205. IEEE, New York (2015).
72. C. Aishwarya, J. Solomon, “Semi-Supervised Learning for Image Classification”, International Journal of Research Publication and Reviews, Vol (5), Issue (6), June (2024), Page –11-17
73. A. Guilherme, O. Laio, F.Stefano,C. Viviana, and D.Leandro, “ Video-Based Human Activity Recognition Using Deep Learning Approaches”, Sensors 2023, 23, 6384.

09 October 2026 04:30:20

74. H. Liu, C. Liu ,and R. Ding, “Semi-supervised long short-term memory for human action recognition”, The Journal of Engineering ,The 3rd Asian Conference on Artificial Intelligence Technology (ACAIT 2019), 2020, Vol. 2020 Iss. 13, pp. 373-378.
75. Bi. Haixia, P. Miquel, S.Raul ,F. Peter, AND C.Ian,”An active semi supervised deep learning model for human activity recognition “,Journal of Ambient Intelligence& Humanized Computing (2023) 14:13049–13065
76. X. Xiao, D.Xu, W.Wan, “Overview: Video recognition from handcrafted method to deep learning method”, Proceedings of theInternational Conference on Audio, Language and Image Processing, Shanghai, China, 11–12 July 2016; pp. 646–651
77. M. Hssayeni,S.Saxena, R.Ptucha, A.Savakis, “Distracted driver detection: Deep learning vs handcrafted features”. Electron. Imaging 2017, 2017, 20–26.
78. M. Zare, D.Alebiosu,S.Lee, “Comparison of handcrafted features and deep learning in classification of medical x-ray images”, Proceedings of the Fourth International Conference on Information Retrieval and Knowledge Management (CAMP), Kota Kinabalu, Malaysia, 26–28 March 2018; pp. 1–5.
79. T. Georgiou, Y.Liu, W.Chen, M.Lew, “A survey of traditional and deep learning-based feature descriptors for high dimensional data in computer vision”, Int. J. Multimed. Inform. Retrieval 2019, 1–36.
80. K. Simonyan, A.Zisserman,”Two-stream convolutional networks for action recognition in videos”, Proceedings of the Advances in Neural Information Processing Systems, Montreal, Canada, 8–13 December 2014; pp. 568–576.
81. L. Wang, Y.Qiao, X.Tang,”Action recognition with trajectory-pooled deep-convolutional descriptors”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Boston, MA, USA, 7–12 June 2015; pp. 4305–4314.
82. C.Feichtenhofer, A.Pinz, A.Zisserman, A, “Convolutional two-stream network fusion for video action recognition”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 27–30 June 2016; pp. 1933–1941.

09 October 2026 04:30:20